

Information about Lily

Academic Background

- **Status:** B.S. candidate in Data Science & Psychology at Northeastern University (Class of 2026)
- **GPA:** 3.89 / 4.00
- **Honors:** University Honors Program, scholarship recipient, Dean's List
- **Availability:** Full-time co-op from July–December 2025

Skills & Tools

- **Languages:** Python, SQL, R, Java, HTML, CSS, JavaScript
- **Libraries/Frameworks:** Pandas, Seaborn, Scikit-learn, BeautifulSoup, Selenium, SciPy, Requests, Plotly, Altair
- **Tools:** Tableau, Git, Google Cloud Platform, AWS
- **OS:** Windows, Ubuntu Linux

Extracurriculars

- NEU Women in Technology
- Concert Band
- Explore Program Mentor & Ambassador

Professional Experience

- **UKG - Data Science Practice Intern**
 - Developing dynamic calendar features & AI tool customization for BDR Playbook
- **UKG - Marketing Operations Intern**

- Automated newsletter unsubscribe process (Python), reducing time from 5 min to 10 sec per URL
- **Northeastern University - Teaching Assistant**
 - *Database Design*: SQL support, grading, office hours for 100+ students
 - *Foundations of Data Science*
- **Artistic Affairs - Social Media Manager**
 - Produced 10 weekly videos, managed 3 companies' online presence

Projects

- **Unsubscribe Assistant**: Python + Streamlit tool automating email unsubscription (97% faster)
- **Baseball Dashboard**: Historical MLB data analytics using Lahman's Dataset (1880–2015); optimized pipelines from CSV to database tables
- **RAG-volution**: Retrieval-Augmented Generation chatbot querying DS4300 notes using LLMs

RAG-volution Details

- **Purpose**: AI chatbot to answer course-related queries with high retrieval accuracy
- **Languages/Tools**: Python, Docker
- **Functions**:
 - PDF → JSON ingestion
 - Embedding + indexing
 - Retrieval + LLM response generation
- **Vector DBs**: Redis, ChromaDB, Qdrant
- **Embedding Models**:

- `all-MiniLM-L6-v2`
 - `all-mpnet-base-v2`
 - `mxbai-embed-large`
- **LLMs Used:** Ollama3:2, Mistral
- **Optimization Techniques:**
 - Chunk sizes (200, 500, 1000 tokens)
 - Overlaps (0, 50, 100 tokens)
- **Example Queries:**
 - “What are the ACID components?”
 - “Write a MongoDB query...”
 - “Difference between B+ tree and AVL tree?”
- **Setup:**
 - Redis (port 6379) and Qdrant (port 6333) in Docker
 - Scalable and domain-adaptable

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